Overview

The Australian Animal Disease Genomics Initiative is a national program to improve how Australia detects, manages and understands animal diseases through genomics.

Bioplatforms Australia established this initiative with the animal health sector. It has $1 million in funding. This support will produce core molecular data and methods for national surveillance, diagnostics, and disease intervention across livestock, aquaculture, and wildlife systems.

Initiative Lead: Sarah Richmond, General Manager – Science

Initiative Chair: Prof. Mark Hutchinson

Project Contact: srichmond@bioplatforms.com

Project Status: Active

Objective alignment:

- 2 - Improve Australia's surveillance and diagnostic capacity for animal pests and diseases

Activity alignment:

- 2.2 - Develop and implement novel technologies, such as POC animal testing and genomics, to address gaps in diagnostic capacity

Project updates

Future Actions

Following the closure of the recent call, successful projects will be assessed and announced in June 2026. Selected teams will then commence data generation through participating national facilities, with resulting datasets curated and made available through appropriate repositories and national data platforms.

The Initiative will continue to expand the Reference Genome Library and Animal Virome Atlas, building a coordinated national resource that supports animal health laboratories, biosecurity agencies, researchers and industry partners in detecting, tracking and responding to animal diseases.

Collaborations are welcomed at any time. Contact the Initiative lead for further information.

May 2026

The Australian Animal Disease Genomics Initiative ran a second request for project partnership proposals, which closed on 1 May 2026. Information on the request is available on the Bioplatforms Australia website.

This round invited proposals to generate foundational genomic data for priority animal pathogens, parasites, and viromes across livestock, wildlife, and environmentally important species.

Approved projects will be supported through the Bioplatforms Australia laboratory network to generate high-quality reference genomes, population genomic datasets, and metagenomic data. These datasets will expand Australia's national pathogen genome resources and improve capacity for disease surveillance, diagnostics, and early detection of emerging threats affecting animal health, agriculture, and ecosystems.

September 2025

Successful projects were announced. Find further details at Bioplatforms Australia.

Under the Reference Genome Library, projects are producing high-quality genomes for priority pathogens and parasites. These include Coxiella burnetii (Q fever), Moraxella, Vibrio, Trypanosoma, the brown stomach worm, and Marteilia sydneyi (QX disease). These datasets will improve diagnostics, strain typing, outbreak tracking, and future vaccine or treatment work.

Under the Animal Virome Atlas, projects are mapping viral diversity in honeybees, native stingless bees, ticks, lizards, koalas, marsupials, and wildlife hospitals. They study how viruses move between wildlife, livestock, and people. They also study how stress, landscape and climatic change shape disease risk.

These projects unite universities, state agencies, wildlife hospitals, and industry partners. Together, they will build a stronger national picture of animal pathogens. They expand our capacity for genomic surveillance, and give reference laboratories and agencies better tools to act early and protect Australia's animals, industries, and ecosystems.

July 2025

The Initiative launched an open Request for Partnership. It accepted proposals for projects to generate foundational genomic data for priority animal pathogens, parasites, and viromes.

The call focused on two themes:

- Reference Genome Library – Build a collection of high-quality, openly available genomes for key animal pathogens. These include viruses, bacteria, fungi, protozoa, and parasites.
- Animal Virome Atlas – Characterise viral diversity and ecology across animal populations and environments.

Data types supported through this round include:

- long-read and short-read genomes
- population genomics (via whole-genome resequencing)
- transcriptomics
- metagenomic sequencing (for Animal Virome Atlas theme)
- proteomics and metabolomics may also be supported where they enhance outcomes.

See more

See more about the Bioplatforms Australia project.